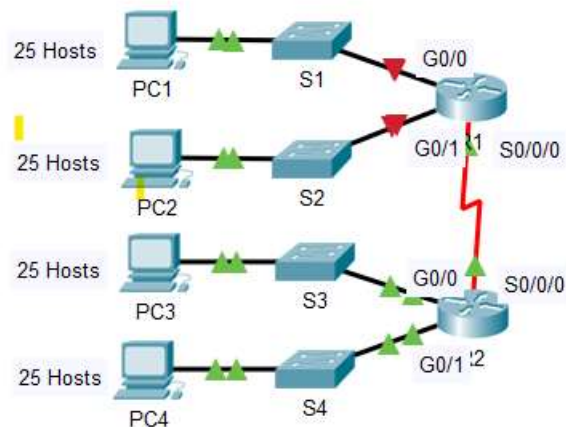


Lab5- Subnetting Scenario

Chapter 8 Activity

Consider the following topology



Resources

Packet Tracer

File needed: Lab5-SubnettingScenario.pka

Objectives

Part 1: Design an IP Addressing Scheme

Part 2: Assign IP Addresses to Network Devices and Verify Connectivity

Scenario

In this activity, you are given the network address of 192.168.100.0/24 to subnet and provide the IP addressing for the network shown in the topology. Each LAN in the network requires enough space for, at least, 25 addresses for end devices, the switch and the router. The connection between R1 to R2 will require an IP address for each end of the link.

Part 1: Design an IP Addressing Scheme

Step 1: Subnet the 192.168.100.0/24 network into the appropriate number of subnets.

- Based on the topology, how many subnets are needed?
- How many bits must be borrowed to support the number of subnets in the topology table?
- How many subnets does this create?
- How many usable hosts does this create per subnet?

Note: If your answer is less than the 25 hosts required, then you borrowed too many bits.

- Calculate the binary value for the first five subnets. The first subnet is already shown.

Net 0: 192 . 168 . 100 . 0 0 0 0 0 0 0 0

Net 1: 192 . 168 . 100 . ____ ____ ____ ____ ____ ____ ____ ____

Net 2: 192 . 168 . 100 . ____ ____ ____ ____ ____ ____ ____ ____

Net 3: 192 . 168 . 100 . ____ ____ ____ ____ ____ ____ ____ ____

Net 4: 192 . 168 . 100 . ____ ____ ____ ____ ____ ____ ____ ____

- Calculate the binary and decimal value of the new subnet mask.

11111111.11111111.11111111. ____ ____ ____ ____ ____ ____ ____ ____

255 . 255 . 255 . _____

- Fill in the **Subnet Table**, listing the decimal value of all available subnets, the first and last usable host address, and the broadcast address. Repeat until all addresses are listed.

Note: You may not need to use all rows.

Subnet Table

Subnet Number	Subnet Address	First Usable Host Address	Last Usable Host Address	Broadcast Address
0				
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Step 2: Assign the subnets to the network shown in the topology.

- Assign Subnet 0 to the LAN connected to the GigabitEthernet 0/0 interface of R1:
- Assign Subnet 1 to the LAN connected to the GigabitEthernet 0/1 interface of R1:
- Assign Subnet 2 to the LAN connected to the GigabitEthernet 0/0 interface of R2:
- Assign Subnet 3 to the LAN connected to the GigabitEthernet 0/1 interface of R2:
- Assign Subnet 4 to the WAN link between R1 to R2:

Step 3: Document the addressing scheme.

Fill in the **Subnet Table** using the following guidelines:

- Assign the first usable IP addresses to R1 for the two LAN links and the WAN link.
- Assign the first usable IP addresses to R2 for the LANs links. Assign the last usable IP address for the WAN link.
- Assign the second usable IP addresses to the switches.
- Assign the last usable IP addresses to the hosts.
- Fill in the addressing table

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0			
	G0/1			
	S0/0/0			
R2	G0/0			
	G0/1			
	S0/0/0			
S1	VLAN 1			
S2	VLAN 1			
S3	VLAN 1			
S4	VLAN 1			
PC1	NIC			
PC2	NIC			
PC3	NIC			
PC4	NIC			

Part 2: Assign IP Addresses to Network Devices and Verify Connectivity

Most of the IP addressing is already configured on this network. Implement the following steps to complete the addressing configuration.

Step 1: Configure IP addressing on R1 LAN interfaces.

Step 2: Configure IP addressing on S3, including the default gateway.

Step 3: Configure IP addressing on PC4, including the default gateway.

Step 4: Verify connectivity.

You can only verify connectivity from R1, S3, and PC4. However, you should be able to ping every IP address listed in the **Addressing Table**.